# LOCI: An Egocentric-Allocentric Dual Memory for Robotic Spatiotemporal Reasoning

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***Abstract*— Autonomous mobile robots operating over long horizons must remember what they observed, where and when they observed it, and which visual evidence supports their answers. Existing memory systems often prioritize semantic retrieval over visual histories or metric structure through 3-D scene graphs, leaving persistent object identity, temporal context, and fine-grained visual evidence insufficiently integrated. We introduce LOCI, a dual-memory framework that organizes robot experience into persistent object-centric 3-D entities and temporally localized episodes. Object memory consolidates identity and metric geometry across observations, while episodic memory captures scene context, activities, and observed changes. Observation links connect entities to their supporting encounters, enabling an agent to combine spatial and temporal retrieval with targeted visual inspection when textual summaries are insufficient. We evaluate LOCI on LMEE-Bench and OC-NaVQA, covering simulated indoor and real-world outdoor environments. LOCI improves question-answering accuracy on the LMEE-Bench 58-task subset and full benchmark by 8.7% and 10.6%, respectively, and reduces spatial and temporal localization errors on OC-NaVQA by 14.9% and 53.6% relative to the most competitive baselines. Deployment on a Unitree G1 humanoid further demonstrates long-horizon memory retrieval and evidence-grounded reasoning in a real-world office environment.**

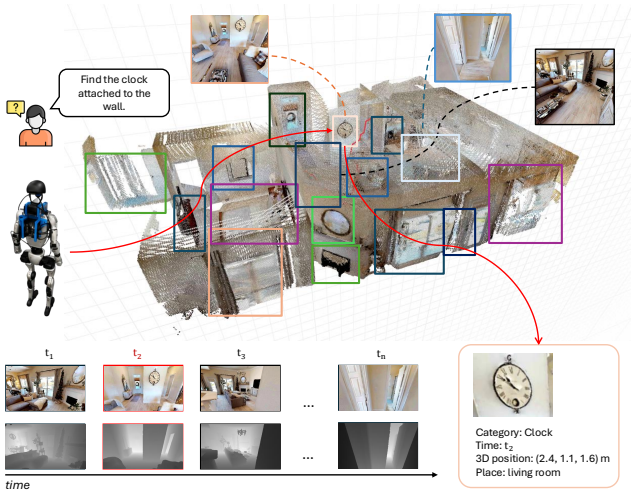



Fig. 1. LOCI organizes robot experience into persistent object-centric 3-D memory and episodic observation intervals. Observation links connect mapped entities to the context and visual evidence of individual encounters. A multimodal agent retrieves from both memories and inspects supporting evidence when semantic summaries are insufficient.

## I. Introduction

Autonomous robots operating over extended periods must reason about more than their current surroundings. They must recall previously encountered objects, recover where and when those encounters occurred, and interpret changes in the environment [1]–[4]. Consider the question: *"Where did I last see the laptop near the coffee machine, and what was placed next to it?"* Answering it requires connecting an object's identity and location with the scene observed during a particular encounter. As visual histories grow, a robot must retrieve this evidence selectively while preserving the details needed to support its answer.

Existing approaches offer two complementary ways to organize long-horizon experience. Observation-centric systems retain keyframes, captions, or multimodal embeddings and retrieve observations relevant to a query [1], [3], [5], [6]. This organization provides direct access to past encounters while limiting the history presented to the reasoning model. However, object identity and metric geometry may remain implicit in separate observations, requiring the system to recover correspondences across retrieved views. Object-centric maps and 3-D scene graphs instead consolidate observations into persistent entities with semantic and geometric attributes [7]–[11]. They provide a structured basis for spatial reasoning, but an object's consolidated representation alone does not describe everything that happened around it or preserve every visual detail of its encounters.

The central challenge is therefore to connect information accumulated across encounters with information specific to each encounter. Persistent object memory captures identity and metric structure across observations; episodic memory preserves temporally localized scene and event context. Neither semantic descriptions nor consolidated object attributes necessarily retain details that become relevant only when a later question is asked. A useful memory must consequently support retrieval across both abstractions and preserve access to their underlying visual evidence.

We introduce **LOCI**, a dual-memory framework for long-horizon robotic question answering and spatiotemporal grounding. From posed RGB-D observations, LOCI incrementally builds persistent object-centric 3-D entities alongside episodic observation intervals. Object memory consolidates repeated observations into entities with metric geometry and semantic descriptors. Episodic memory summarizes scene context, activities, and observed changes within each interval while retaining sampled visual observations. Explicit observation links connect entities to the intervals in which they appear, allowing object-level information to be recovered in the context of a particular encounter.

Project page and video demonstrations are available at `https://loci-memory.github.io`

A multimodal agent uses these memories to gather evidence for a natural-language question. It retrieves entities for object-level semantic and spatial grounding, searches episodes for temporal and contextual information, and inspects supporting images when additional visual detail is required. This organization allows the agent to move from compact semantic representations to source evidence as needed, without repeatedly processing the complete observation history.

We evaluate LOCI on LMEE-Bench [6] and OC-NaVQA [11], covering simulated indoor environments and real-world outdoor robot trajectories. The evaluation measures question-answering accuracy alongside spatial and temporal grounding, distinguishing semantic answer quality from the recovery of precise locations and times. We further investigate the role of memory organization and evidence retrieval through component ablations, measuring their effects on question-answering accuracy and spatial and temporal grounding. Deployment on a Unitree G1 humanoid further demonstrated memory retrieval and evidence-grounded reasoning in an office environment.

Our main contributions are:

- **A dual-memory representation.** LOCI organizes long-horizon robot experience into persistent object-centric 3-D entities and independently retrievable episodic intervals, retaining metric object information and temporally localized scene context.
- **Retrieval grounded in observation evidence.** Explicit links between entities and their encounters enable a multimodal agent to combine spatial and temporal evidence and inspect supporting images when semantic summaries are insufficient.
- **Evaluation across simulated and physical environments.** Experiments on LMEE-Bench and OC-NaVQA, component ablations, and deployment on a Unitree G1 examine LOCI's performance in long-horizon question answering and spatiotemporal grounding.

## II. Related Work

### A. Long-Horizon Episodic and Retrieval-Based Robotic Memory

Retrieval-based robotic memories organize past experience into compact representations and expose query-relevant evidence to a reasoning model. ReMEmbR [1] stores visual descriptions with metric poses and timestamps for long-horizon retrieval and reasoning. However, caption-based retrieval can overlook visual details that were not considered salient during memory construction. RAVEN [5] addresses this limitation through direct visual embeddings associated with pose and time, reducing dependence on textual summaries. MemoryExplorer [6] instead focuses on memory access, using reinforcement fine-tuning to improve active exploration through memory tools. Related work also investigates structured memory for long egocentric videos [2], [12]

These approaches preserve temporally localized experience, but observation retrieval does not by itself consolidate repeated encounters into persistent 3-D object identities. Although observations may carry spatial information, recovering an object's metric state across viewpoints can still require resolving correspondence among retrieved observations. This motivates maintaining object-level geometry and identity alongside episodic context.

### B. Object-Centric 3-D Spatial Memory and Scene Graphs

Early 3D scene graphs organize metric-semantic environments into objects, places, rooms, and buildings [7], [13]–[15]. Open-vocabulary approaches extend these representations with language-aligned semantics. ConceptGraphs [8], [16], [17] associates and fuses object observations across posed RGB-D frames, while HOV-SG [9] organizes open-vocabulary features across objects, rooms, and floors. These representations support persistent spatial grounding, but their object and place abstractions do not directly represent the scene and event context of individual encounters.

Subsequent work expands the information and reasoning supported by scene graphs. Open3DSG [18] predicts open-set object relationships, addressing relational semantics rather than episodic context. GraphEQA [10] combines hierarchical scene graphs with visual observations for embodied question answering, while SayPlan [19] selectively retrieves task-relevant subgraphs for planning. These methods demonstrate that visual evidence and selective retrieval can complement structured spatial representations. Khronos [20] and DAAAM [11] further preserve timestamped description histories within a 4-D scene graph, supporting long-horizon spatiotemporal reasoning. STaR [21] constructs a task-agnostic multimodal memory that captures object attributes, spatial relations, and dynamic events. Its task-conditioned retrieval uses the information bottleneck principle to select compact, relevant evidence while reducing redundancy

Nevertheless, histories attached to mapped entities provide a different abstraction from independently retrievable intervals that summarize scene-wide activities and context. LOCI builds on this distinction by factoring experience into persistent object memory and episodic interval memory. The former consolidates identity and metric geometry across observations; the latter preserves scene and event context within temporal intervals. Observation links connect entities with their encounters, allowing a multimodal agent to combine evidence from both abstractions and recover fine-grained visual details when semantic summaries are insufficient.

## III. Problem Formulation

We consider a robot that accumulates a history of posed RGB-D observations.

$$\mathcal{M}_{1:T} = \{o_t\}_{t=1}^{T}, \qquad o_t = (I_t, D_t, \mathbf{T}_t^{wc}, \tau_t), \qquad (1)$$

where $I_t$ and $D_t$ denote the RGB image and metric depth, $\mathbf{T}_t^{wc} \in SE(3)$ is the camera pose in the world frame, and $\tau_t$ is the capture time.. Given a natural-language query $Q$ about the robot's accumulated experience, the system must

produce an evidence-grounded answer $A$. Ideally, the answer is inferred from the complete observation history:

$$A^\star = \arg\max_A p(A \mid Q, \mathcal{M}_{1:T}), \tag{2}$$

but repeatedly exposing $\mathcal{M}_{1:T}$ to a multimodal model is neither scalable nor robust to irrelevant observations. We instead construct a persistent memory graph $\mathcal{H}$ and retrieve a compact evidence set $\mathcal{R}(Q)$.

$$p(A \mid Q, \mathcal{M}_{1:T}) \approx p(A \mid Q, \mathcal{R}(Q)), \quad \mathcal{R}(Q) \subseteq \mathcal{H} \cup \mathcal{M}_{1:T}. \tag{3}$$

The objective is therefore to retrieve a compact set of evidence sufficient to answer the query, rather than merely returning the most semantically similar memory items.

This formulation presents three core challenges:

**Task-agnostic memory construction.** How can the robot incrementally transform a growing multimodal history into a persistent and extensible representation that preserves semantic, spatial, temporal, and fine-grained visual information for previously unseen queries?

**Task-conditioned retrieval.** Given a completed memory $\mathcal{M}_{1:T}$ and a query $Q$, how can the robot efficiently retrieve a compact, relevant, and non-redundant evidence set $\mathcal{R}(Q)$ while maintaining scalability with respect to deployment duration and memory size?

**Evidence-grounded reasoning.** How can the robot jointly reason over structured memory and its supporting visual observations to produce accurate answers across different semantic, spatial, and temporal granularities?

Overall, the objective is to construct a scalable long-term memory that preserves both compact structured knowledge and observation-level visual evidence, and that dynamically exposes the appropriate evidence for each query. This enables embodied robots to answer open-ended questions about their completed experience without repeatedly processing the full sensor history.

## IV. Methodology

LOCI organizes posed RGB-D observations into two complementary memories: an object memory $\mathcal{G}$ that consolidates persistent identities, metric geometry, and semantics across viewpoints, and an episodic memory $\mathcal{D}$ that summarizes scene and event context within temporal intervals. Observation links $\mathcal{E}$ connect entities with the intervals in which they were observed.

We denote the complete memory as

$$\mathcal{H} = (\mathcal{G}, \mathcal{D}, \mathcal{E}), \tag{4}$$

Given a question, a multimodal agent selectively retrieves from both memories and accesses supporting visual evidence when additional detail is needed. Figure 2 summarizes the framework.

### A. Object-Centric 3-D Memory

*a) 2-D Perception and 3-D Lifting:* For each RGB keyframe $I_t$, a vision-language model (VLM) generates object descriptions from the observed scene. A promptable segmentation model uses these descriptions to extract instance masks $\{m_{t,i}\}_{i=1}^{M_t}$. We extract a normalized DINOv2 [22] appearance descriptor $\mathbf{f}_{t,i}$ from each masked crop and back-project the corresponding depth pixels using camera intrinsics. The camera pose transforms the resulting points into the world frame, yielding an object observation $P_{t,i}$. Memory construction uses only the observation history and does not depend on future questions.

*b) Object Association and Fusion.:* For each observation $(P_{t,i}, \mathbf{f}_{t,i})$, a spatial index retrieves candidate entities whose expanded axis-aligned bounding boxes overlap the observation bounds. We score each candidate $j$ using geometric support and visual similarity:

$$\phi(i,j) = \phi_{\text{geo}}(i,j) + \phi_{\text{vis}}(i,j). \tag{5}$$

Geometric support measures the fraction of observed points that lie within a distance $\delta_{\text{nn}}$ of the entity's accumulated point cloud $P_j$:

$$\phi_{\text{geo}}(i,j) = \frac{1}{|P_{t,i}|} \sum_{\mathbf{p} \in P_{t,i}} \mathbb{I}\left[\min_{\mathbf{q} \in P_j} \|\mathbf{p} - \mathbf{q}\|_2 < \delta_{\text{nn}}\right]. \tag{6}$$

Each entity maintains a descriptor bank $\mathcal{B}_j$ containing up to $K$ descriptors from its highest-quality observed views. Visual similarity is the maximum rescaled cosine similarity between the incoming descriptor and this bank:

$$\phi_{\text{vis}}(i,j) = \max_{\mathbf{b} \in \mathcal{B}_j} \frac{1 + \cos(\mathbf{f}_{t,i}, \mathbf{b})}{2}. \tag{7}$$

The observation is assigned to the highest-scoring candidate if its score exceeds the association threshold $\gamma_{\text{assoc}}$. Otherwise, a new entity is initialized. For a matched entity, we merge the observed points into its point cloud and voxel-downsample the result using voxel size $\delta_v$. The entity's bounding box is updated from the fused geometry.

*c) View Selection and Semantic Enrichment:* To favor views in which an object is large and centrally located, we score observation $k$ of entity $j$ as

$$q_{j,k}^{\text{view}} = \sqrt{a_{j,k}} \left(\frac{1 + c_{j,k}}{2}\right), \tag{8}$$

where $a_{j,k} \in [0,1]$ is the mask area normalized by image area and $c_{j,k} \in [0,1]$ measures proximity to the image center, with larger values indicating more central observations.

A VLM combines selected high-quality views into an entity description, which is encoded for semantic retrieval. We also extract a SigLIP [23] embedding to support text-image retrieval.

### B. Episodic Memory

To retain scene context beyond individual objects, we divide the observation history into intervals of duration $\Delta t_{\text{mem}}$. For each nonempty interval, a VLM summarizes

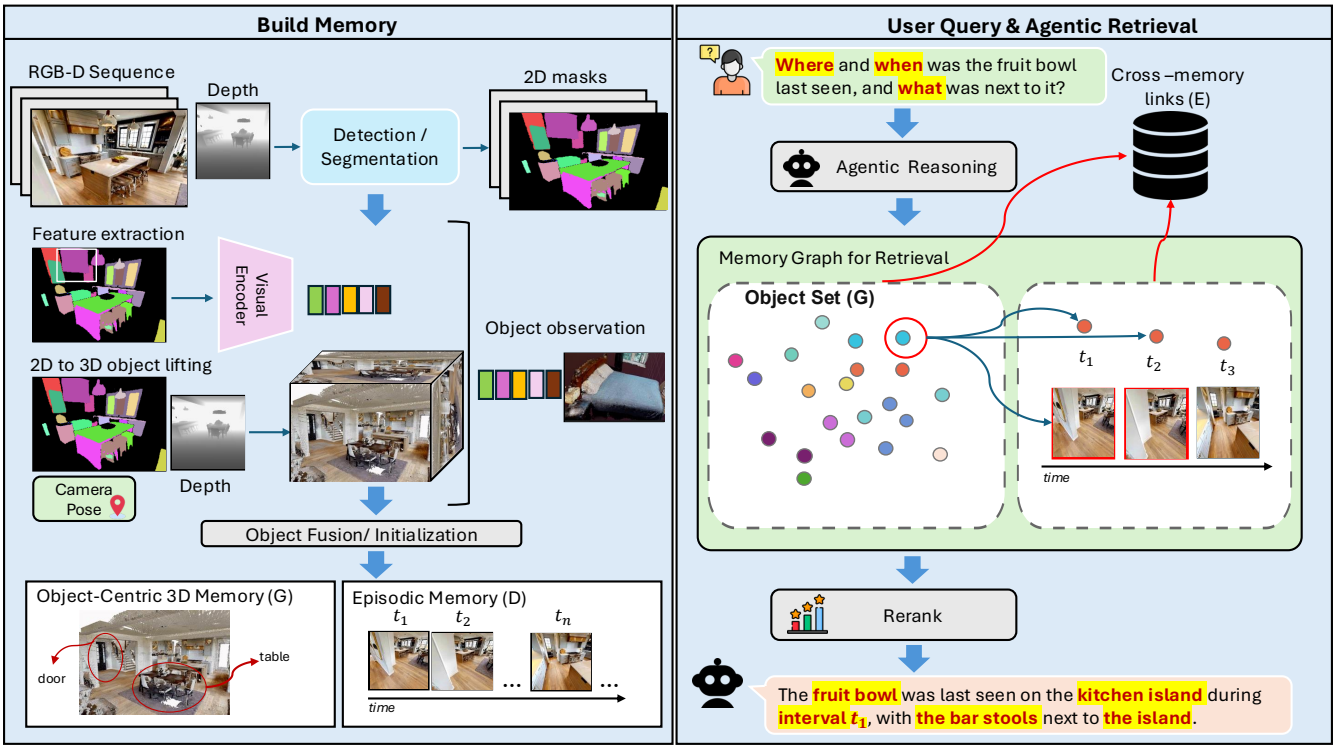



Fig. 2. Overview of LOCI. Object memory consolidates persistent entities across observations, while episodic memory retains scene and event context within temporal intervals. Observation links connect the two memories, enabling selective retrieval of object information, encounter-specific context, and supporting visual evidence.

the ordered visual observations, describing scene context, activities, and observed changes. Each interval stores an identifier, its time span, a summary, a text embedding, and RGB-D keyframes sampled at a controlled rate.

The summary supports semantic retrieval, while the retained visual evidence allows the agent to examine details omitted during summarization. Observation links $\mathcal{E}$ associate entities with the intervals in which they were observed, retaining the corresponding observation references and view-quality scores. These links preserve the connection between an object's aggregated representation and its individual encounters.

For example, after identifying an emergency call pole through object retrieval, the agent can request evidence from a specified encounter and inspect the surrounding scene. The object memory supplies persistent identity and metric grounding, while the associated interval supplies the context of that encounter.

### C. Evidence-Grounded Agentic Reasoning

Given a question $Q$, a multimodal agent iteratively selects memory tools based on the question and the evidence retrieved so far. Tool responses are accumulated into an evidence set $\mathcal{R}$, which informs subsequent retrieval and the final answer.

*a) Memory Access.:* The agent uses four complementary tools:

- **Entity retrieval** searches $\mathcal{G}$ using text-embedding similarity over entity descriptions and SigLIP text–image similarity over entity visual embeddings. It returns entity identifiers, descriptions, 3-D positions, and bounding boxes.
- **Episodic retrieval** searches $\mathcal{D}$ using the two-stage procedure below and returns interval identifiers, time spans, summaries, and relevance scores.
- **Entity-view retrieval** follows observation links for a specified entity and returns supporting visual evidence and object masks ranked by view quality. A temporal constraint restricts retrieval to the relevant encounters.
- **Interval inspection** retrieves the retained visual evidence for a specified interval, allowing the agent to examine scene context and details omitted from its summary.

These tools allow the agent to combine persistent object information with temporally localized context. An entity result can guide the selection of a relevant encounter, while an episodic result can provide context for subsequent object queries. Retrieval of compact descriptions and geometry is separated from visual inspection, allowing the agent to request additional visual evidence when needed to resolve ambiguity or examine fine-grained details.

*b) Two-Stage Episodic Retrieval:* For each episodic query, we retrieve $K_{\text{coarse}}$ candidate intervals using cosine similarity between query and caption embeddings. A cross-encoder then scores each query–caption pair, and the $K_{\text{fine}}$ highest-scoring intervals are returned to the agent, where

$K_{\text{fine}} << K_{\text{coarse}}$.

*c) Answer Generation:* The agent continues retrieval until it judges the evidence sufficient or reaches a maximum of $M_{\max}$ tool calls. It then generates an answer $\hat{A}$ conditioned on the question and accumulated evidence $\mathcal{R}$. The aim is to approximate the full-history answer $A^\star$ defined in Eq. (2), using selectively retrieved object information, episodic context, and visual evidence.

### D. Implementation Details

*a) Perception and Memory Construction.:* We use Qwen3-VL-8B-instruct [24] to generate object descriptions and SAM3 [25] for promptable segmentation. Appearance descriptors are extracted using DINOv2 ViT-B/14 [22], while entity visual embeddings use siglip-base-patch16-224. Entity descriptions and episodic summaries are encoded with Qwen3-Embedding-8b. The VLM receives up to 5 high-quality views when generating an entity description. Episodic intervals have duration $\Delta t_{\text{mem}} = 4$ s, and supporting keyframes are retained at 1 Hz.

*b) Association and View Retention.:* We set the geometric matching distance to $\delta_{\text{nn}} = 0.1$ m and the association threshold to $\gamma_{\text{assoc}} = 1.10$. The fused point clouds are downsampled with voxel size $\delta_v = 0.05$ m. Candidate bounding boxes are expanded by 0.15 m before overlap testing. Each entity's descriptor bank retains at most $K = 10$ normalized DINOv2 descriptors, selected by the view-quality score in Eq. (8).

*c) Retrieval and Reasoning.:* We use Qwen3-Reranker-0.6B to rerank $K_{\text{coarse}} = 20$ episodic candidates and return the top $K_{\text{fine}} = 3$. The reasoning agent uses GPT-4o [26] with a maximum of $M_{\max} = 12$ tool calls per question.

## V. Experiments & Results

We evaluate LOCI on LMEE-Bench [6], a simulated indoor benchmark, and OC-NaVQA [11], a real-world long-horizon benchmark. The experiments examine question answering, spatial and temporal grounding, and the contributions of individual memory components.

Baseline Implementation: We reproduce FARM, DAAAM, and STaR under our evaluation protocol. All reproduced methods are evaluated on the same trajectory observations, poses, query sets, and reasoning backbone. We retain each method's original memory representation and retrieval procedure . On LMEE-Bench, we additionally include the original paper's reported subset results for its GPT-4o retrieval-augmented QA baseline and MemoryExplorer.

### A. Evaluation on LMEE-Bench

LMEE-Bench is constructed from HM3D-Sem to evaluate long-term memory during multi-goal embodied exploration. The full split contains 166 trajectories, 828 navigation goals, and 406 questions. The randomly sampled subset contains 58 trajectories, 272 goals, and 145 questions. Questions cover attribute, counting, location, relationship, and state reasoning across easy, medium, and hard difficulty levels.

We evaluate multiple-choice QA by replaying complete RGB-D trajectories with ground-truth poses and timestamps. Each method constructs its memory incrementally, and all associated questions are issued after exploration ends. This protocol evaluates memory-based QA given recorded experience, rather than online navigation performance.

TABLE I

Multiple-choice QA accuracy (%) on LMEE-Bench. FARM, DAAAM, STaR, and LOCI use GPT-4o for reasoning. Numbers in parentheses indicate the number of questions in each difficulty category.

| Method | Easy | Medium | Hard | Total |
|---|---|---|---|---|
| *Subset* | *(35)* | *(94)* | *(16)* | *(145)* |
| GPT-4o | 65.7 | 51.1 | 56.3 | 55.2 |
| MemoryExplorer | 68.6 | 63.8 | 68.7 | 65.5 |
| DAAAM | 57.1 | 59.6 | 62.5 | 59.1 |
| STaR | 68.6 | 71.3 | **75.0** | 71.0 |
| FARM | 77.1 | 69.2 | 43.8 | 68.3 |
| **LOCI** | **85.7** | **76.6** | 62.5 | **77.2** |
| *Full* | *(80)* | *(272)* | *(54)* | *(406)* |
| MemoryExplorer | 66.2 | 64.3 | **64.8** | 64.8 |
| DAAAM | 52.5 | 48.2 | 61.1 | 50.7 |
| STaR | 62.5 | 56.3 | 63.0 | 58.4 |
| FARM | 73.8 | 64.0 | 59.3 | 65.3 |
| **LOCI** | **78.8** | **72.1** | 63.0 | **72.2** |

LOCI achieves the highest overall accuracy among the evaluated methods on both splits (Table I), reaching **77.2%** on the subset and **72.2%** on the full benchmark. These results exceed the strongest reproduced baselines by **8.7%** and **10.6%**, respectively. The advantage holds with GPT-4o shared across LOCI and the reproduced baselines, supporting the effectiveness of the proposed memory construction and retrieval system.

For reference, LOCI also exceeds the published subset results for GPT-4o and MemoryExplorer by approximately **40%** and **17.9%**, respectively.

LOCI ranks first on the easy and medium tiers of both splits. Performance on the hard tier is more mixed; however, these tiers contain only 16 and 54 questions, respectively. As a result, one or two examples can substantially affect the reported percentages, making the estimates sensitive to question composition and individual failure cases.

### B. Evaluation on OC-NaVQA

OC-NaVQA [11] extends NaVQA [1], which is based on CODa [27] robot navigation recordings, by annotating spatial queries with the actual 3D positions of referenced objects and correcting inaccurate labels. It replaces question-specific observation windows with complete sequences, reaching up to 35.8 minutes and 1.64 km of robot travel. The benchmark includes spatial, temporal, binary, and descriptive questions. We report question accuracy, positional error in metres, and temporal error in minutes.

Each sequence is processed continuously to construct a single memory, which is frozen before the associated

questions are issued. We exclude four of the 210 questions because their provided ground-truth observation time does not fall into the memory window, leaving 206 questions. The same exclusions are applied to all compared methods.

LOCI maintains competitive QA accuracy while achieving the lowest positional and temporal errors among the evaluated methods (Table II). The principal advantage is in grounding. LOCI achieves a positional error of **28.28 m**, a **14.9%** reduction relative to STaR, the strongest spatial baseline, and a **46.3%** reduction relative to DAAAM. Its temporal error of **1.47 min** is less than half that of DAAAM, the strongest temporal baseline at 3.17 min, representing a **53.6%** reduction.

### C. Ablation Study on OC-NaVQA

We ablate the 3D scene graph, episodic memory retrieval, episodic reranking, and SigLIP-based visual retrieval to examine the contribution of each component (Table III). The complete LOCI system achieves the best performance across all three metrics, indicating that its spatial, temporal, and visual retrieval components provide complementary benefits.

*a) Complementary Roles of the Two Memories.:* The two memory components exhibit distinct and complementary effects. Removing the 3-D scene graph increases positional error from **28.28 m** to **51.68 m**, the largest spatial degradation among all ablations. In contrast, removing episodic memory retrieval causes the largest drop in question accuracy, from **69.70%** to **61.11%**, while also increasing temporal error from **1.47 min** to **2.78 min**. These results reflect the different roles of the two memories: the object-centric 3-D representation provides persistent metric grounding, whereas episodic memory preserves the temporal and contextual information needed for question answering. Their combination is therefore important for jointly recovering what was observed, where it occurred, and when it was encountered.

*b) Reranking and Visual Retrieval.:* The retrieval mechanisms further improve the quality of the evidence exposed to the reasoning agent. Without episodic reranking, positional error increases from 28.28 m to **44.33 m**, while temporal error rises to 1.71 min and question accuracy decreases to 68.42%. This suggests that reranking helps distinguish semantically similar episodes and recover evidence from the appropriate spatiotemporal context.

Removing SigLIP-based retrieval reduces question accuracy from 69.70% to **67.68%**, a decrease of **2.02 percentage points**, and increases positional and temporal errors to 33.75 m and 2.00 min, respectively. This indicates that visual similarity complements description-based retrieval when textual representations alone are insufficient. Overall, the ablations show that LOCI benefits not only from its dual-memory representation, but also from retrieval mechanisms that effectively expose the relevant evidence stored within it.

### D. Real-World Deployment on a Humanoid Robot

We deploy LOCI on a Unitree G1 humanoid to evaluate the framework in a physical indoor environment. The robot is equipped with RGB-D sensing and an onboard NVIDIA Jetson AGX Orin for localization, navigation, and low-level system integration. Computationally intensive perception and memory processing are executed on an external workstation equipped with an NVIDIA GeForce RTX 5090 GPU and an Intel Core Ultra 9 285K CPU. The robot and workstation communicate over a local Ethernet network through ROS 2.

During deployment, RGB-D observations and robot pose estimates are streamed to the workstation, where LOCI incrementally updates its persistent object-centric 3-D memory and episodic memory. Given a natural-language query, the reasoning agent retrieves relevant object, temporal, and visual evidence from the accumulated experience. For queries that specify or imply a navigation target, the corresponding grounded 3-D position is returned to the robot and forwarded to Nav2 for path planning and execution.

This architecture separates high-level perception and long-horizon memory reasoning from real-time navigation and control: memory construction and retrieval are performed offboard, while localization, planning, and motion execution remain onboard. The resulting system enables LOCI to connect natural-language queries to grounded observations and use the retrieved spatial evidence for subsequent robot navigation.

## VI. Conclusion

### A. Summary

In this work, we presented LOCI, a dual-memory retrieval and reasoning framework for long-horizon embodied question answering. LOCI combines a persistent object-centric 3-D scene representation $\mathcal{G}$ with an episodic memory $\mathcal{D}$ of temporally indexed captions and visual evidence. The two representations provide complementary abstractions: persistent entities support metric spatial grounding, while episodic memory preserves transient scene and event context that may not be captured by object-level state alone.

### B. Limitations and Future Work

LOCI has four main limitations:

- **Perception and geometric uncertainty.** Errors in segmentation, depth, and camera pose can compromise object geometry and association, particularly in large outdoor environments. The current representation does not explicitly model these uncertainties. Uncertainty-aware association and probabilistic entity states could improve robustness to ambiguous observations.
- **Identity maintenance in changing scenes.** Persistent entities are best suited to static or slowly changing objects. Object motion and visually similar instances can cause incorrect associations across encounters. Explicit state histories and change detection could help distinguish object identity from changes in location or appearance.
- **Information loss in episodic retrieval.** Details omitted from episode summaries may be difficult to retrieve even when supporting images remain available. Richer multimodal indexing could improve evidence recall,

TABLE II

RESULTS ON OC-NAVQA. ALL METHODS USE GPT-4O FOR REASONING. BOLD INDICATES THE BEST RESULT.

| Method | Question Accuracy [%] ↑ | Positional Error [m] ↓ | Temporal Error [min] ↓ |
|---|---|---|---|
| FARM | 70.70 | 59.22 | 6.46 |
| DAAAM | 64.90 | 52.67 | 3.17 |
| STaR | **72.73** | 33.24 | 3.24 |
| **LOCI** | 69.70 | **28.28** | **1.47** |

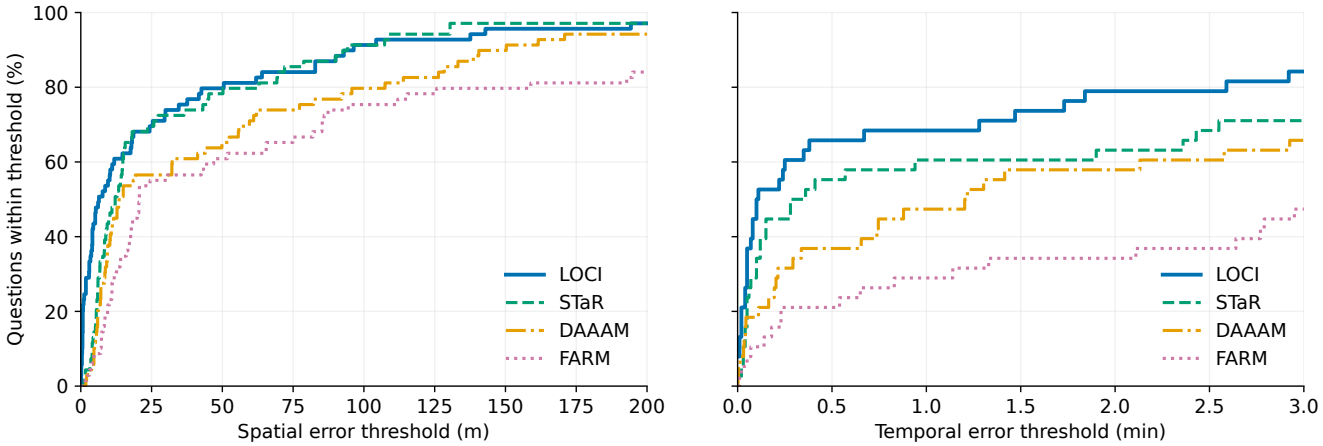



Fig. 3. Cumulative error distributions for spatial and temporal grounding on OC-NaVQA. LOCI exhibits lower errors across a larger fraction of queries.

TABLE III

ABLATION STUDY OF LOCI ON THE OC-NAVQA BENCHMARK.

| Variant | Accuracy (%) ↑ | Position (m) ↓ | Time (min) ↓ |
|---|---|---|---|
| **LOCI** | **69.70** | **28.28** | **1.47** |
| w/o object-centric 3D memory | 67.68 | 51.68 | 2.16 |
| w/o episodic memory retrieval | 61.11 | 38.88 | 2.78 |
| w/o reranker | 68.42 | 44.33 | 1.71 |
| w/o SigLIP | 67.68 | 33.75 | 2.00 |

while adaptive sampling could retain observations based on novelty or uncertainty rather than a fixed schedule.

- **Dependence on the reasoning model.** Tool selection, evidence integration, and answer generation rely on a general-purpose multimodal model, making accuracy and latency sensitive to the chosen backbone. Learned retrieval policies could improve evidence selection and reduce unnecessary tool calls.

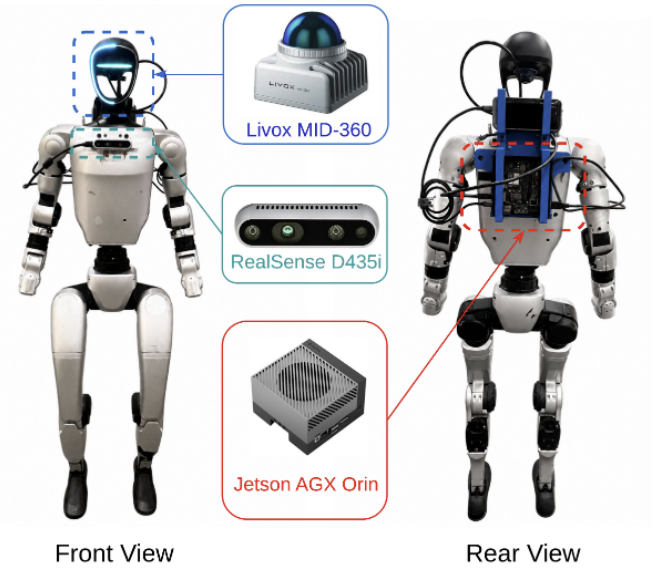



Fig. 4. Unitree G1 platform and hardware configuration used for the real-world deployment.

## REFERENCES


[1] A. Anwar, J. Welsh, J. Biswas, S. Pouya, and Y. Chang, "Remembr: Building and reasoning over long-horizon spatio-temporal memory for robot navigation," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 2838–2845.

[2] G. Goletto, T. Nagarajan, G. Averta, and D. Damen, "Amego: Active memory from long egocentric videos," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024.

[3] Q. Xie, S. Y. Min, P. Ji, Y. Yang, T. Zhang, K. Xu, A. Bajaj, R. Salakhutdinov, M. Johnson-Roberson, and Y. Bisk, "Embodied-rag: General non-parametric embodied memory for retrieval and generation," *arXiv preprint arXiv:2409.18313*, 2024.

[4] A. Majumdar, A. Ajay, X. Zhang, P. Putta, S. Yenamandra, M. Henaff, S. Silwal, P. Mcvay, O. Maksymets, S. Arnaud, K. Yadav, Q. Li, B. Newman, M. Sharma, V. Berges, S. Zhang, P. Agrawal, Y. Bisk, D. Batra, M. Kalakrishnan, F. Meier, C. Paxton, S. Sax, and A. Rajeswaran, "Openeqa: Embodied question answering in the era of foundation models," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[5] Y. Hu, Z. Zheng, L. Zha, C. Xing, R. Singh, O. Hossain, A. Loquercio, and D. Shah, "Raven: Long-horizon reasoning & navigation with a visuo-spatio-temporal memory," *arXiv preprint arXiv:2606.25206*,

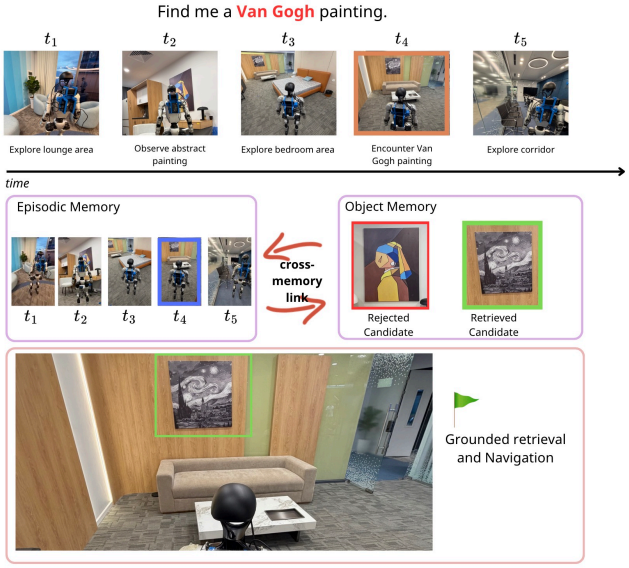



Fig. 5. Real-world deployment of LOCI. The robot accumulates observations during exploration, retrieves complementary evidence from episodic and object-centric memories, and grounds the query to a navigation target in the environment.


2026.

[6] S. Wang, B. Liu, Z. Gao, L. Ma, X. Wang, Y. Xie, and X. Tan, "Explore with long-term memory: A benchmark and multimodal llm-based reinforcement learning framework for embodied exploration," in *Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)*, 2026.

[7] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial perception system for 3D scene graph construction and optimization," 2022.

[8] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa *et al.*, "Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 5021–5028.

[9] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, "Hierarchical open-vocabulary 3d scene graphs for language-grounded robot navigation," in *First Workshop on Vision-Language Models for Navigation and Manipulation at ICRA 2024*, 2024.

[10] S. Saxena, B. Buchanan, C. Paxton, P. Liu, B. Chen, N. Vaskevicius, L. Palmieri, J. Francis, and O. Kroemer, "Grapheqa: Using 3d semantic scene graphs for real-time embodied question answering," in *Conference on Robot Learning (CoRL)*, 2025.

[11] N. Gorlo, L. Schmid, and L. Carlone, "Describe anything anywhere at any moment," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 35 002–35 013.

[12] Y. Yang, H. Yang, J. Zhou, P. Chen, H. Zhang, Y. Du, and C. Gan, "3d-mem: 3d scene memory for embodied exploration and reasoning," in *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, June 2025, pp. 17 294–17 303.

[13] I. Armeni, Z.-Y. He, J. Gwak, A. R. Zamir, M. Fischer, J. Malik, and S. Savarese, "3d scene graph: A structure for unified semantics, 3d space, and camera," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 5664–5673.

[14] A. Rosinol, A. Gupta, M. Abate, J. Shi, and L. Carlone, "3d dynamic scene graphs: Actionable spatial perception with places, objects, and humans," in *Robotics: Science and Systems (RSS)*, 2020.

[15] A. Rosinol, A. Violette, M. Abate, N. Hughes, Y. Chang, J. Shi, A. Gupta, and L. Carlone, "Kimera: From slam to spatial perception with 3d dynamic scene graphs," *The International Journal of Robotics Research*, vol. 40, no. 12–14, pp. 1510–1546, 2021.

[16] K. Jatavallabhula, A. Kuwajerwala, Q. Gu, M. Omama, T. Chen, S. Li, G. Iyer, S. Saryazdi, N. Keetha, A. Tewari, J. Tenenbaum, C. de Melo, M. Krishna, L. Paull, F. Shkurti, and A. Torralba, "Conceptfusion: Open-set multimodal 3d mapping," *Robotics: Science and Systems (RSS)*, 2023.

[17] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, London, UK, 2023.

[18] S. Koch, N. Vaskevicius, M. Colosi, P. Hermosilla, and T. Ropinski, "Open3dsg: Open-vocabulary 3d scene graphs from point clouds with queryable objects and open-set relationships," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 14 183–14 193.

[19] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Sünderhauf, "Sayplan: Grounding large language models using 3d scene graphs for scalable robot task planning," in *Conference on Robot Learning (CoRL)*. PMLR, 2023, pp. 23–72.

[20] L. Schmid, M. Abate, Y. Chang, and L. Carlone, "Khronos: A unified approach for spatio-temporal metric-semantic slam in dynamic environments," in *Proc. of Robotics: Science and Systems (RSS)*, Delft, Netherlands, July 2024.

[21] M. Yuan, H. Zhang, M. Mohammadi, R. Li, J. Shan, and S. L. Waslander, "Star: Scalable task-conditioned retrieval for long-horizon multi-modal robot memory," *IEEE Robotics and Automation Letters*, vol. 11, no. 5, pp. 5994–6001, 2026.

[22] M. Oquab, T. Darcet, T. Moutakanni, H. V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, R. Howes, P.-Y. Huang, H. Xu, V. Sharma, S.-W. Li, W. Galuba, M. Rabbat, M. Assran, N. Ballas, G. Synnaeve, I. Misra, H. Jegou, J. Mairal, P. Labatut, A. Joulin, and P. Bojanowski, "Dinov2: Learning robust visual features without supervision," *arXiv:2304.07193*, 2023.

[23] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," 2023. [Online]. Available: https://arxiv.org/abs/2303.15343

[24] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv, C. Zheng, D. Liu, F. Zhou, F. Huang, F. Hu, H. Ge, H. Wei, H. Lin, J. Tang, J. Yang, J. Tu, J. Zhang, J. Yang, J. Yang, J. Zhou, J. Zhou, J. Lin, K. Dang, K. Bao, K. Yang, L. Yu, L. Deng, M. Li, M. Xue, M. Li, P. Zhang, P. Wang, Q. Zhu, R. Men, R. Gao, S. Liu, S. Luo, T. Li, T. Tang, W. Yin, X. Ren, X. Wang, X. Zhang, X. Ren, Y. Fan, Y. Su, Y. Zhang, Y. Zhang, Y. Wan, Y. Liu, Z. Wang, Z. Cui, Z. Zhang, Z. Zhou, and Z. Qiu, "Qwen3 technical report," 2025. [Online]. Available: https://arxiv.org/abs/2505.09388

[25] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, C. Ryali, K. V. Alwala, H. Khedr, A. Huang, J. Lei, T. Ma, B. Guo, A. Kalla, M. Marks, J. Greer, M. Wang, P. Sun, R. Rädle, T. Afouras, E. Mavroudi, K. Xu, T.-H. Wu, Y. Zhou, L. Momeni, R. Hazra, S. Ding, S. Vaze, F. Porcher, F. Li, S. Li, A. Kamath, H. K. Cheng, P. Dollár, N. Ravi, K. Saenko, P. Zhang, and C. Feichtenhofer, "Sam 3: Segment anything with concepts," 2025. [Online]. Available: https://arxiv.org/abs/2511.16719

[26] OpenAI, "Gpt-4o system card," *arXiv preprint arXiv:2410.21276*, 2024.

[27] A. Zhang, C. Eranki, C. Zhang, J.-H. Park, R. Hong, P. Kalyani, L. Kalyanaraman, A. Gamare, A. Bagad, M. Esteva *et al.*, "Towards robust robot 3d perception in urban environments: The ut campus object dataset," *arXiv preprint arXiv:2309.13549*, 2023.